

LAPORAN PRAKTIKUM

STRUKTUR DATA

MODUL [No. 01]

CODE BLOCKS IDE & PENGENALAN BAHASA C++



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PROGRAM STUDI S1 INFORMATIKA

FAKULTAS INFORMATIKA

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LATIHAN SOAL MODUL

Link Repo Github Modul X : (masukan link folder Repo Github KHUSUS MODUL INI)

1. Soal 1

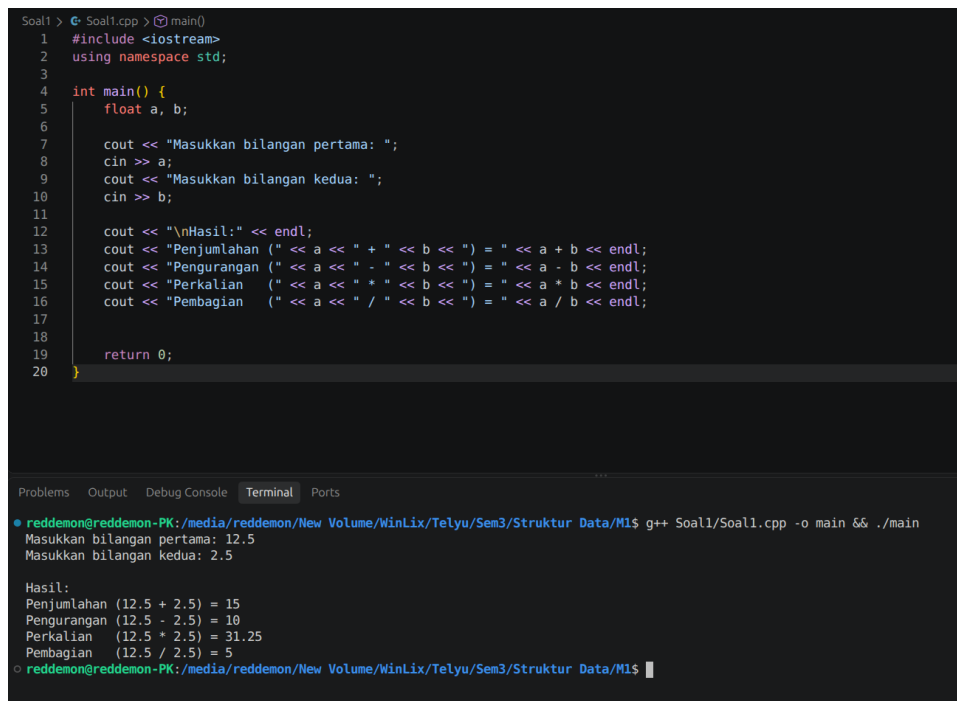
Source Code

```
#include <iostream>
using namespace std;
int main() {
    float a, b;

    cout << "Masukkan bilangan pertama: ";
    cin >> a;
    cout << "Masukkan bilangan kedua: ";
    cin >> b;

    cout << "\nHasil:" << endl;
    cout << "Penjumlahan (" << a << " + " << b << ") = " << a
+ b << endl;
    cout << "Pengurangan (" << a << " - " << b << ") = " << a
- b << endl;
    cout << "Perkalian (" << a << " * " << b << ") = " << a
* b << endl;
    cout << "Pembagian (" << a << " / " << b << ") = " << a
/ b << endl;
    return 0;
}
```

Screenshoot program



```
Soal1 > g++ Soal1.cpp > ./main()
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      float a, b;
6
7      cout << "Masukkan bilangan pertama: ";
8      cin >> a;
9      cout << "Masukkan bilangan kedua: ";
10     cin >> b;
11
12     cout << "\nHasil:" << endl;
13     cout << "Penjumlahan (" << a << " + " << b << ") = " << a + b << endl;
14     cout << "Pengurangan (" << a << " - " << b << ") = " << a - b << endl;
15     cout << "Perkalian (" << a << " * " << b << ") = " << a * b << endl;
16     cout << "Pembagian (" << a << " / " << b << ") = " << a / b << endl;
17
18     return 0;
19 }
20

Problems Output Debug Console Terminal Ports
● reddemon@reddemon-PK: /media/reddemon/New Volume/WinLix/Telyu/Sem3/Struktur Data/M1$ g++ Soal1/Soal1.cpp -o main && ./main
Masukkan bilangan pertama: 12.5
Masukkan bilangan kedua: 2.5

Hasil:
Penjumlahan (12.5 + 2.5) = 15
Pengurangan (12.5 - 2.5) = 10
Perkalian (12.5 * 2.5) = 31.25
Pembagian (12.5 / 2.5) = 5
● reddemon@reddemon-PK: /media/reddemon/New Volume/WinLix/Telyu/Sem3/Struktur Data/M1$
```

Deskripsi program

Program diatas bekerja dengan membaca dua masukan bertipe float dari pengguna, kemudian langsung melakukan kalkulasi dan menampilkan hasil penjumlahan, pengurangan, serta perkalian, sekaligus menyertakan pengecekan kondisi agar pembagian dengan angka nol tidak mengeksekusi operasi ilegal.

2. Soal 2

Source Code

```
#include <iostream>
#include <string>
using namespace std;

string satuan[] = {"", "Satu", "Dua", "Tiga", "Empat",
                  "Lima", "Enam", "Tujuh", "Delapan", "Sembilan", "Sepuluh",
                  "Sebelas"};

string konversiKeTeks(int n) {
    if (n == 0) return "Nol";
    if (n <= 11) return satuan[n];
    if (n < 20) return satuan[n - 10] + " Belas";
    if (n < 100) {
        string hasil = satuan[n / 10] + " Puluh";
        if (n % 10 != 0) {
            hasil += " " + satuan[n % 10];
        }
        return hasil;
    }
    if (n == 100) return "Seratus";

    return "Di luar jangkauan (0 - 100)";
}

int main() {
    int angka;
    cout << "Masukkan angka (0 - 100): ";
    cin >> angka;

    if (angka >= 0 && angka <= 100) {
        cout << angka << " : " << konversiKeTeks(angka) <<
endl;
    } else {
        cout << "Input harus antara 0 sampai 100." << endl;
    }

    return 0;
}
```

```
#include <iostream>
using namespace std;

int main() {
    int n;
    cout << "input: ";
    cin >> n;

    cout << "output:\n" << endl;

    for (int i = n; i >= 1; i--) {

        for (int s = 0; s < (n - i) * 2; s++) {
            cout << " ";
        }
    }
}
```

```

        for (int j = i; j >= 1; j--) {
            cout << j << " ";
        }

        cout << "*";

        for (int j = 1; j <= i; j++) {
            cout << " " << j;
        }

        cout << endl;
    }
    for (int s = 0; s < n * 2; s++) {
        cout << " ";
    }
    cout << "*" << endl;

    return 0;
}

```

Screenshoot program

The screenshot shows a C++ program being executed in a terminal. The program's logic is as follows:

- It includes `<iostream>` and uses the `std` namespace.
- In the `main` function, it reads an integer `n` from the user (input: 3).
- It prints "output:" followed by a newline.
- It enters a loop for `i` from `n` down to 1:
 - For each `i`, it prints numbers from `i` down to 1, followed by an asterisk.
 - Then, it prints numbers from 1 up to `i`, separated by spaces.

The output of the program for `n=3` is:

```

3 2 1 * 1 2 3
2 1 * 1 2
1 * 1
*

```

Deskripsi program

Program diatas bekerja dengan menggunakan teknik perulangan bersarang (nested loop) untuk membentuk pola mirror terbalik, di mana baris luar mengatur penurunan tingkat dari \$N\$ ke 1, sementara perulangan di dalamnya mencetak spasi pendorong di kiri, deret angka menurun, satu karakter bintang (*) tepat di tengah, deret angka menaik, dan diakhiri dengan satu bintang tunggal di titik paling bawah.